NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

SECOND YEAR (Computer Science)

SPRING SEMESTER EXAMINATIONS 2024

Batch 2022

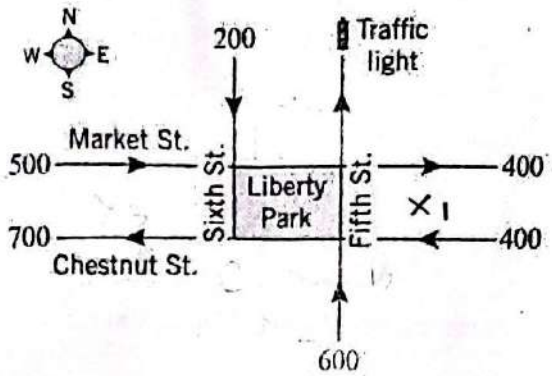
Time : 3 Hours | **Dated : 27-JUL-24** | **Max Marks : 60**

Linear Algebra & Geometry - MT-272

Sir Syed Ashad Ali

Instructions:
1): Attempt all questions.
2): All questions carry equal marks.
3): Assume any missing data.

Q#1 [CLO 1] The **network** in Figure shows a proposed plan for the traffic flow around a new park that will house the Liberty Bell in Philadelphia, Pennsylvania. The plan calls for a **computerized traffic light** at the north exit on Fifth Street, and the diagram indicates the average number of vehicles per hour that are expected to flow in and out of the streets that border the complex. All streets are one-way. **(12)**



(a): How many **vehicles per hour** should the traffic light let through to ensure that the average number of vehicles per hour flowing into the complex is the same as the average number of vehicles flowing out?

(b): Assuming that the traffic light has been set to balance the total flow in and out of the complex, what can you say about the **average number of vehicles per hour** that will flow along the streets that border the complex?

Q#2 [CLO 1] A company produces **Web design, software, and networking** services. View the company as an open economy described by the accompanying table, where input is in dollars needed for $1.00 of output. **(06)**

Input Required per Dollar Output

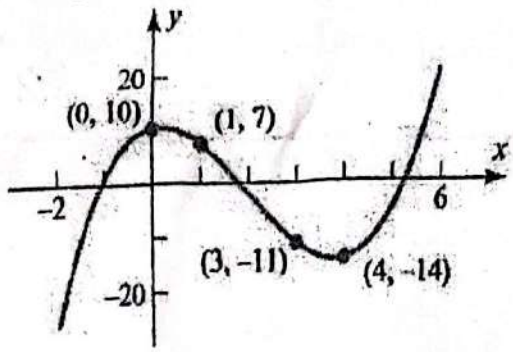
Provider	Web Design	Software	Networking
Web Design	$ 0.40	$ 0.20	$ 0.45
Software	$ 0.30	$ 0.35	$ 0.30
Networking	$ 0.15	$ 0.10	$ 0.20

(a): Find the **consumption matrix** for the company.

(b): Suppose that the customers (the open sector) have a demand for $5400 worth of Web design, $2700 worth of software, and $900 worth of networking. Use row reduction to find a **production vector** that will meet this demand exactly.

CamScanner

(b): Find the **coefficients a, b, c, and d** so that the curve shown in the accompanying figure is the graph of the equation $y = ax^3 + bx^2 + cx + d$. **(06)**



Q#3 [CLO 2] (a): For the given matrix A, **(06)**

$$A = \begin{bmatrix} 2 & -1 & 2 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$$

i) Verify **Cayley Hamilton Theorem.**
ii) Find A^{-1}.
iii) Find A^4.

(b): Consider the **bases** $B = \{u_1, u_2, u_3\}$ and $B = \{u_1', u_2', u_3'\}$ for R^3, where **(06)**

$$u_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, u_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, u_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, u_1' = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, u_2' = \begin{bmatrix} 0 \\ -1 \\ 2 \end{bmatrix}, u_3' = \begin{bmatrix} 2 \\ 3 \\ -5 \end{bmatrix}$$

i. Find the **transition matrix** from B to B'.
ii. Find coordinates of (1, 2, -1) with respect to B'.

Q#4 [CLO 2] For the given matrix A, **(12)**

$$A = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}$$

i) Write **characteristic equation.**
ii) Find **Eigen Values.**
iii) Find **Eigen Vectors.**
iv) Find the **Modal matrix P.**
v) Find the **Diagonal matrix D.**
vi) Find A^3.

Q#5 [CLO 1] (a): (i): Derive the relation between Direction Cosines. **(06)**
(ii): Find the **direction ratios, direction cosines** and **direction angles** of the vector $v = (3, 4, -6)$.

(b): Using the **vector form of equation of a straight line**, identify whether the points $(-7, 5), (-1, 2)$ & $(5, -1)$ are collinear or not. **(06)**

CamScanner

Seat No. CR-22014

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

SECOND YEAR (ARTIFICIAL INTELLIGENCE/ DATA SCIENCE/ CYBER SECURITY)
SPRING SEMESTER EXAMINATIONS 2024
Batch - 2022

Time: 3 Hours | **Dated:27-07-2024**
Max.Marks:60

LINEAR ALGEBRA & GEOMETRY - MT-272

Note: Attempt ALL questions.

Q1.(a)	The matrix A id defined as: $$A = \begin{bmatrix} 1 & 2 & -3 \\ 0 & 3 & 2 \\ 0 & 0 & -2 \end{bmatrix}$$ Find the eigen values of $3A^3 + 5A^2 - 6A + 2I$	(CLO-2)	(12)
Q1.(b)	Find the Euclidean inner product of the vectors $u = (0,0,1)$ and $v = (0,2,2)$ if the angle between them is 45°	(CLO-1)	(3)
Q2.(a)	Find the distance between the point $(3,-1,2)$ and the plane $2x + y - z = 4$. Also reduce the equation into normal form and find its distance from the origin.	(CLO-1)	(6)
Q2.(b)	Find vector and parametric equations for the line in $\mathbb{R}^3$ that passes through the points $P(5,-3,4)$ *and* $Q(1,-2,7)$.	(CLO-1)	(6)
Q2.(c)	Find the norm of vectors a) $v = (-7,2,-1)$ b) $v = (-5,0,1)$	(CLO-1)	(3)
Q3.(a)	Show that $S = \{v_1, v_2\}$ is basis for all vectors in $\mathbb{R}^2$ when $v_1 = (1,1)$ and $v_2 = (1,-1)$	(CLO-2)	(9)
Q3.(b)	Determine whether the vectors: $v_1 = (1,2,-3)^T, v_2 = (1,-3,2)^T, v_3 = (2,-1,5)^T$ are linearly independent or linearly dependent in $\mathbb{R}^3$.	(CLO-1)	(6)
Q4.(a)	Balance the given chemical equation using Gauss Jordan method. $(CH_4) + (O_2) \rightarrow (CO_2) + (H_2O)$	(CLO-2)	(12)
Q4.(b)	Write the names of applications of linear algebra in computer science/real-life.	(CLO-1)	(3)

CamScanner

Ms. Razia Shaheen

SEAT NO. _____

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY
SECOND YEAR (BS IN COMPUTER SCIENCE & INFORMATION TECHNOLOGY)
SPRING SEMESTER EXAMINATIONS 2023
Batch 2021

Time : 3 Hours

Dated : 08-AUG-23
Max Marks : 60

Linear Algebra and Geometry – MT-272

Instructions: Attempt all **five** questions. All questions carry equal marks.

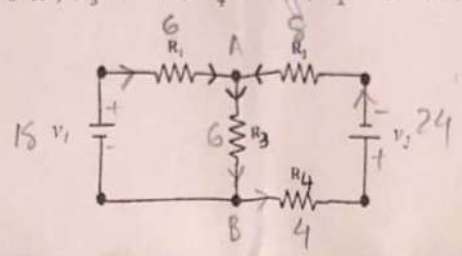
QUESTION 1 CLO 1

(a) Write a balance chemical equation for the following chemical reaction using system of linear equation. (6)

$$C_3H_8 + O_2 \longrightarrow CO_2 + H_2O \quad (propane\ combustion)$$

(b) Analyze the circuit in following figure and find the unknown currents, where

$$R_1 = 6\ \Omega\ ,\ R_2 = 8\ \Omega\ ,\ R_3 = 6\ \Omega\ ,\ R_4 = 4\ \Omega\ ,\ V_1 = 18\ volts \quad V_2 = 24\ volts$$ (6)



QUESTION 2 CLO 1

(a) **Sketch** the image of a triangle with vertices (3,1) , (3,3) and (5,1) under the composite Transformation in $\mathbb{R}^2$ stated as "reflection about a line $y = x$ followed by rotation through an angle of 90^0 followed by a dilation with scaling factor 2 ." (6)

(b) In the given table 1 each plaintext and ciphertext letter assigned the numerical value that specifies its position in the standard alphabet. Decode the message SFANEFZWJH given that it is a Hill Cipher with enciphering matrix $\begin{bmatrix} 4 & 3 \\ 1 & 2 \end{bmatrix}$ and **translate** into plaintext letter. To find the reciprocal, use table 2. (6)

Table 1

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	0

Table 2 Reciprocals Modulo 26

a	1	3	5	7	9	11	15	17	19	21	23	25
a^{-1}	1	9	21	15	3	19	7	23	11	5	17	25

QUESTION 3 CLO 2

(a) Check whether the set of polynomials $\{1 + x\ ,\ 1 - x\ ,\ 1 - x^2\ ,\ 1 - x^3\}$ form a basis for P_3 (cubic polynomial). (6)

(b) Find the Matrix for which the Eigen vectors are $\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$, $\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ *and* $\begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}$ corresponding to Eigen values $\lambda = 2\ ,2$ and 1 respectively. {Hints. Use concept of diagonalization. Find inverse of a matrix by elementary row operations. For multiplication of matrices you can use calculator.} (6)

PTO

Scanned with CamScanner

QUESTION 4 **CLO 2**

Let L_1 and L_2 be the lines whose parametric equations are (12)

$$L_1 : x = 4t \ , \ y = 1 - 2t \ , \ z = 2 + 2t \qquad L_2 : x = 1 + t \ , \ y = 1 - t \ , \ z = -1 + 4t$$

(a) Find the point of intersection of the lines L_1 and L_2 .

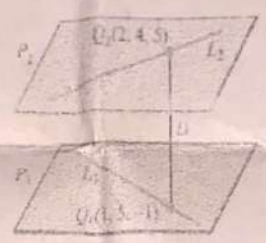
(b) Find, to the nearest degree, the acute angle between L_1 and L_2 at their intersection.

(c) Find parametric equations for the line that is perpendicular to L_1 and L_2 and passes through their point of intersection.

QUESTION 5 CLO 2

(a) Develop an equation of the plane that contains the line $x = -2 + 3t$, $y = 4 + 2t$, $z = 3 - t$ and perpendicular to the plane $x - 2y + z = 5$. (6)

(b) The lines $L_1 : x = 1 + 4t$, $y = 5 - 4t$, $z = -1 + 5t$, $L_2 : x = 2 + 8t$, $y = 4 - 3t$, $z = 5 + t$ are skew. Find the distance between them. (6)



****

Scanned with CamScanner

SEAT NO. _____

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

SECOND YEAR(BACHELOR OF SCIENCE IN COMPUTER SCIENCE & INFORMATION TECHNOLOGY)

SPRING SEMESTER EXAMINATIONS 2022

BATCH 2020

Dated:19-08-2022

Time: 3 Hours

Max.Marks:60

Linear Algebra & Geometry - MT-272

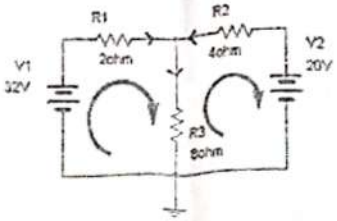
Instructions: Attempt all five questions. All questions carry equal marks.

QUESTION 1 CLO 1

(a) Write a balance chemical equation for the following chemical reaction using system of linear equation. (6)

$$KNO_3 + H_2CO_3 \longrightarrow K_2CO_3 + HNO_3$$

(b) Analyze the following circuit and list the current in each branch of the circuit. (6)



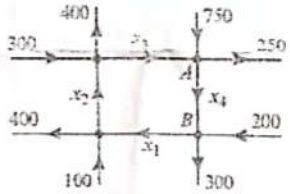
QUESTION 2 CLO 1

(a) The accompanying figure shows a network of one-way streets with traffic flowing in the directions indicated. The flow rates along the streets are measured as the average number of vehicles per hour.

i. Set up a linear system whose solution provides the unknown flow rates.

ii. Solve the system for the unknown flow rates.

iii. If the flow along the road from A to B must be reduced for construction, what is the minimum flow that is required to keep traffic flowing on all roads. (8)



(b) Find the standard matrix for the composite transformation $T: R^3 \rightarrow R^3$ that first rotates a vector counterclockwise about z-axis through an angle 30^0, then reflect the resulting vector about the yz-plane and then project that vector orthogonally onto xy-plane. (4)

QUESTION 3 CLO 2

(a) Find the vector component of $v = (2, -1, 3)$ along $a = (4, -1, 2)$ and vector component of u orthogonal to a. (6)

(b) Solve the following system of non-linear equations for the unknown x, y and z. (6)

$$x^2 + y^2 + z^2 = 6$$
$$x^2 - y^2 + 2z^2 = 2$$
$$2x^2 + y^2 - z^2 = 3$$

QUESTION 4 CLO 2

(a) Check whether the set of vectors $\{(1,2,2,-1), (4,9,9,-4), (5,8,9,-5)\}$ in $\mathcal{R}^4$ are linearly independent. (6)

(b) List the Eigen values and Eigen vectors for matrix $A = \begin{bmatrix} 3 & 2 & 4 \\ 2 & 0 & 2 \\ 4 & 2 & 3 \end{bmatrix}$ (6)

QUESTION 5 CLO 2

(a) Develop an equation of the plane passes through the points $A(0,1,1)$, $B(1,1,2)$ and $C(-1,2,-2)$. (6)

(b) Find the shortest distance between the lines $x - 1 = \frac{y-2}{2} = \frac{z-3}{3}$ and $x = 1 - y = \frac{z-2}{-2}$ (6)

Scanned with CamScanner

SEAT NO. _____

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

SECOND YEAR(BACHELOR OF SCIENCE IN
COMPUTER SCIENCE & INFORMATION TECHNOLOGY)
SPRING SEMESTER EXAMINATIONS 2021
BATCH 2019

Time: 3 Hours | **Dated:28-08-2021**
Max.Marks:60

Linear Algebra & Geometry - MT-272

Instructions: Attempt all **five** questions. All questions carry equal marks.

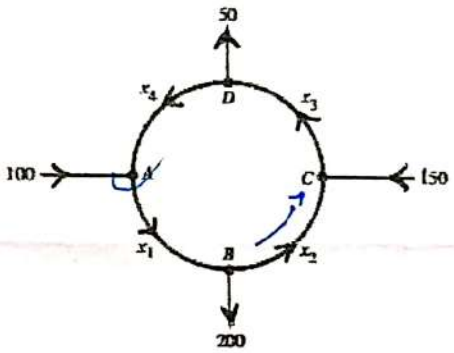
QUESTION 1 CLO 1 **12 marks**

(a) Figure represents the traffic entering and leaving a "roundabout" (rotatory) road junction. Such junctions are very common in our country. Construct a Mathematical model that describes the flow of traffic along the various branches. What is the minimum flow theoretically possible along the branch BC? What are the other flows at that time? . (The units of flow are vehicles per hour.)

$$\begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

$\lambda^3 - 5\lambda^2 + 4\lambda - 3 = 0$

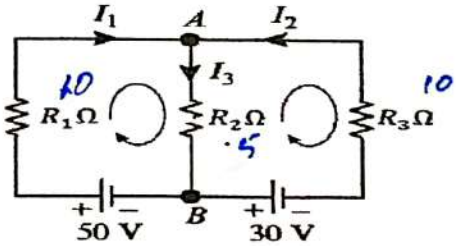
4.220
$+ 0.389$
$- 0.389$



$= 2$
$= 2$
$= 0$
4

$2(2) - 1(1) + 0$
$4 - 1$
3

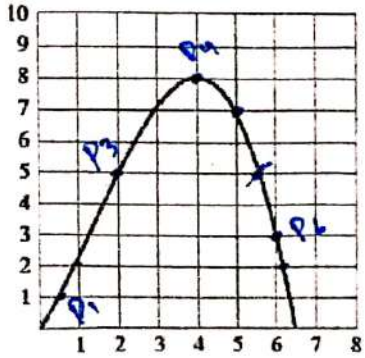
(b) Choose $R_1 = 10$, $R_2 = 5$ *and* $R_3 = 10$. Analyze the following circuit, list the current in each branch of the circuit and find their values.



QUESTION 2 CLO 1 **12marks**

(a) The accompanying figure shows the graph of a cubic polynomial. Find the polynomial

$ax^3 + bx^2 + cx + d = 0$



P.T.O

(b) An 'L' shape has corners at the points with coordinates (1, 4), (1, 7), (2, 7), (2, 5), (3, 5) and (3, 4). Use transformation matrix to draw the reflections of the shape about the lines: (i) $y = 0$ (ii) $x = y$

QUESTION 3 **CLO 2** **12 marks**

(a) Find the Eigen values and corresponding eigen vectors of $A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$.

$0, y$

$(x, 0)$

(b) Solve the following system of non-linear equations for x, y *and* z.

$$x^2 + y^2 + z^2 = 6$$
$$x^2 - y^2 + 2z^2 = 2$$
$$2x^2 + y^2 - z^2 = 3$$

QUESTION 4 **CLO 2** **12 marks**

(a) Find the vector component of u along a and the vector component of u orthogonal to a if $u = (2, -1, 3)$ *and* $a = (4, -1, 2)$.

(b) Examine whether the vector space of given polynomials in x are linearly dependent.
$p(x) = 1 + 2x + 3x^2$, $q(x) = 3 - x + 4x^2$, *and* $r(x) = 4 + x + 7x^2$.

(c) Find the spherical coordinates equation for the sphere $x^2 + y^2 + (z-1)^2 = 1$.

$x = \rho \sin\phi \cos\theta \quad y = \rho$

QUESTION 5 **CLO 2** **12 marks**

(a) Determine whether the given lines are parallel, intersect or are skew. If they intersect find their point of intersection.

$$L_1 : x = 2 + t \quad , \quad y = 2 + 3t \quad , \quad z = 3 + t$$
$$L_2 : x = 2 + t \quad , \quad y = 3 + 4t \quad , \quad z = 4 + 2t$$

(b) Find parametric equations for the line in which the planes $3x - 6y - 2z = 15$ and $2x + y - 2z = 5$ intersect. Find also the angle between the planes.

$-1, \frac{7}{}, $

***

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

SECOND YEAR FALL SEMESTER (ELECTRICAL ENGINEERING)
EXAMINATIONS 2018
BATCH 2017

Dated: 16-02-2019

Time: 3 Hours

Max.Marks: 60

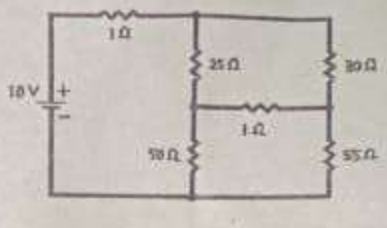
Linear Algebra & Geometry - MT-272

Instructions: Attempt all five questions. All questions carry equal marks.

QUESTION 1 — CLO 1

(a) Find the standard matrix for the linear operator $T : \mathbb{R}^3 \to \mathbb{R}^3$ that first rotates a vector counter clockwise about $z-axis$ through an angle 30^0 then reflects the resulting vector about the $yz-plane$, and then project that vector orthogonally onto the $xy-plane$. (6)

(b) Analyze the following circuit and list the current in each branch of the circuit. (6)



QUESTION 2 — CLO 2

(a) Identify the rank of the matrix $\begin{bmatrix} -1 & 2 & 0 & 4 & 5 & -3 \\ 3 & -7 & 2 & 0 & 1 & 4 \\ 2 & -5 & 2 & 4 & 6 & 1 \\ 4 & -9 & 2 & -4 & -4 & 7 \end{bmatrix}$ (6)

(b) Use Cayley-Hamilton Theorem to find matrix A^{-1} and matrix A^3 when $A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$ (6)

QUESTION 3 — CLO 2

(a) Describe the Quadric surface $x_1^2 - x_3^2 - 4x_1x_2 + 4x_2x_3$ in terms of new variables by eliminating cross product terms. (6)

(b) Solve the following system of non-linear equations for the unknown angles α, β and γ where $0 \le \alpha \le 2\pi$, $0 \le \beta \le 2\pi$, $0 \le \gamma < \pi$. (6)

$$\sin\alpha + 2\cos\beta + 3\tan\gamma = 0$$
$$2\sin\alpha + 5\cos\beta + 3\tan\gamma = 0$$
$$-\sin\alpha - 5\cos\beta + 5\tan\gamma = 0$$

QUESTION 4 — CLO 2

Let L_1 and L_2 be the lines whose parametric equations are

$$L_1 : x = 1 + 2t,\ y = 2 - 3t,\ z = 4t \text{ and } L_2 : x = 17 - 4t,\ y = t,\ z = 2t$$

(a) Show that L_1 and L_2 are skew lines. (4)

(b) Find the shortest distance between them. (4)

(c) Find the parametric equation of their common perpendicular. (4)

QUESTION 5 — CLO 2

(a) Develop an equation of the plane that contains the line $x = -1 + 3t$, $y = 5 + 2t$, $z = 2 - t$ and perpendicular to the plane $2x - 4y + 2z = 9$. (4)

(b) Construct the equation of the sphere whose centre lies on the line $x = y = z$ and passing through the points $(5, 3, 0)$ and $(-1, 4, 1)$. (4)

(c) Model an equation of the surface of revolution obtained by rotating the line $2x + 3y = 6$ in $xy-$plane about (i) $x-axis$ (ii) $y-axis$. (4)